

Solution QUIZ 5 (Dated: 04-11-2024)
Data Structures and Algorithm Analysis

Roll No: _____

Name: _____

Q1. An integer vector has numbers 0 to 9 only. A duplicate removal consists of choosing two adjacent and equal numbers and removing them. We repeatedly make duplicate removals until we no longer can. Return the final list of values after all such duplicate removals have made. Sample input/ output given in the box.

```
vector<int> duplicateRemove(vector<int>& v){
    stack<int> st;
    for (int x : v)
        if (st.empty() || st.top() != x)    st.push(x);
        else
            st.pop();
    vector<int> res(st.size());
    for (int i = st.size() - 1 ; i >= 0 ; i--,st.pop())
        res[i] = st.top();
    return res;
}
```

Input: {3,5,5,3,8,3}
Output: {8,3}
Input: {3,7,5,5,7,4}
Output: {3,4}

Q2. Fill array with first five letters of your first name and run the code. Show complete working and output.

```
void print (char x[], int i, int n){
    if (i >= n) return;
    print (x, i + 1, n);
    cout << x[i] << ' ';
    print (x, i + 2, n);
}
int main(){
    char c[5]={'P', 'A', 'K', 'I', 'S'};
    print(c, 0, 5);
    return 0;
}
```

S I K S A S I P S I K S

```
print(c, 0, 5) -> cout << x[0], next call with i = 2
print(c, 1, 5); -> cout << x[1], next call with i = 3
    print(c, 2, 5); -> cout << x[2], next call with i = 4 to print x[4], S again
        print(c, 3, 5); -> cout << x[3], next call with i = 5 that returns
            print(c, 4, 5); -> cout << x[4], next call with i = 6 that returns
                print(c, 5, 5); -> return

(The output before next call is S I K S A)
        print(c, 3, 5); -> cout << x[3], next call with i = 5 that returns
            print(c, 4, 5); -> cout << x[4], next call with i = 6 that returns
                print(c, 5, 5); -> return

(The output before next call is S I K S A S I)
            print(c, 2, 5); -> cout << x[2], next call with i = 4 to print x[4], S again
                print(c, 3, 5); -> cout << x[3], next call with i = 5 that returns
                    print(c, 4, 5); -> cout << x[4], next call with i = 6 that returns
                        print(c, 5, 5); -> return

(The output before next call is S I K S A S I P S I K)
                    print(c, 4, 5); -> cout << x[4], next call with i = 6 that returns
                        print(c, 5, 5); -> return
```